rose steeply for about the first 150 generations. Thereafter it declined rather gradually, approaching extinction around generation 1,000. Harrington did well temporarily for the same reason as my original cheat did. It exploited softies like Tit for Two Tats (too forgiving) while these were still around. Then, as the softies were driven extinct, Harrington followed them, having no easy prey left. The field was free for 'nice' but 'provocable' strategies like Tit for Tat.

Tit for Tat itself, indeed, came out top in five out of six runs of Round 3, just as it had in Rounds 1 and 2. Five other nice but provocable strategies ended up nearly as successful (frequent in the population) as Tit for Tat; indeed, one of them won the sixth run. When all the nasties had been driven extinct, there was no way in which any of the nice strategies could be distinguished from Tit for Tat or from each other, because they all, being nice, simply played COOPERATE against each other.

A consequence of this indistinguishability is that, although Tit for Tat seems like an ESS, it is strictly not a true ESS. To be an ESS, remember, a strategy must not be invadable, when it is common, by a rare, mutant strategy. Now it is true that Tit for Tat cannot be invaded by any nasty strategy, but another nice strategy is a different matter. As we have just seen, in a population of nice strategies they will all look and behave exactly like one another: they will all COOPERATE all the time. So any other nice strategy, like the totally saintly Always Cooperate, although admittedly it will not enjoy a positive selective advantage over Tit for Tat, can nevertheless drift into the population without being noticed. So technically Tit for Tat is not an ESS.

You might think that since the world stays just as nice, we could as well regard Tit for Tat as an ESS. But alas, look what happens next. Unlike Tit for Tat, Always Cooperate is not stable against invasion by nasty strategies such as Always Defect. Always Defect does well against Always Cooperate, since it gets the high temptation score every time. Nasty strategies like Always Defect will come in to keep down the numbers of too nice strategies like Always Cooperate.

But although Tit for Tat is strictly speaking not a true ESS, it is probably fair to treat some sort of mixture of basically nice but retaliatory 'Tit for Tat-like' strategies as roughly equivalent to an ESS in practice. Such a mixture might include a small admixture of nastiness. Robert Boyd and Jeffrey Lorberbaum, in one of the more interesting follow-ups to Axelrod's work, looked at a mixture of Tit for Two Tats